

	HINDUSTAN PETROLEUM GREEN R&D CENTRE, KIADB INDUSTRIAL AREA, TARABAHALLI VILLAGE, HOSKOTE TALUK, BANGALORE -560067 Ph: 080-28078521/080-28078524 MATERIAL SAFETY DATA SHEET- HP NanOKoat
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SECTION I - PRODUCT IDENTIFICATION	
Chemical Name	HP NanOKoat
Chemical Application	To provide improved corrosion protection

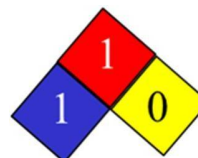
SECTION II - COMPOSITION/INFORMATION ON INGREDIENTS	
INGREDIENT(S)	Wt%
Epoxy resin	55-65%
Hardener	10-15%
Fillers	25-30%
	100

The product has a shelf life of 1 year without addition of hardener if it is stored in the original container between 18 to 25 °C. Also keep it away from humidity and excessive heat. Any concentration shown as a range is to protect the confidentiality or is due to batch variation.

SECTION III - HAZARDS IDENTIFICATION

The product has to be labeled due to the calculation procedure of international guidelines.

R 10 Flammable.
R 20/21 Harmful by inhalation and in contact with skin.
R 36/38 Irritating to eyes and skin.



Health = 1
Fire = 1
Reactivity = 0

NFPA Label

SECTION IV - FIRST AID MEASURES

General information:

Symptoms of poisoning may even occur after several hours; therefore, medical observation for at least 48 hours after the accident.

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After inhalation:

Remove to fresh air. If breathing is difficult, give oxygen. Obtain medical advice if there are persistent symptoms.

After skin contact:

Wash immediately with water and soap and rinse thoroughly.

After eye contact:

Rinse opened eyes for several minutes under running water. If symptoms persist, consult a doctor.

After swallowing:

If swallowed, call a physician immediately. Only induce vomiting at the instruction of a physician. Never give anything by mouth to an unconscious person.

SECTION V - FIRE- FIGHTING MEASURES

Suitable extinguishing agents:

Use water spray or fog, carbon dioxide or dry chemical.

Unsuitable Extinguishing Media:

Full water jet.

Protective equipment:

Firefighters, and others exposed, wear self-contained breathing apparatus.

SECTION VI- ACCIDENTAL RELEASE MEASURES

Person-related safety precautions: Where exposure level is not known, wear approved, positive pressure, self-contained respirator. Where exposure level is known, wear approved respirator suitable for level of exposure.

Measures for environmental protection:

Do not allow product to reach sewage system or any water course.

Inform respective authorities in case of escape into the water course or sewage system.

Measures for cleaning/collecting:

Absorb with liquid-binding material (sand, diatomite, acid binders, universal binders, sawdust). Dispose of contaminated material as waste accordingly. Clean up affected area.

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SECTION VII - HANDLING AND STORAGE

Handling:

Information for safe handling:

Ensure good ventilation/exhaustion at the workplace. Wash hands thoroughly after handling.

Information about protection against explosions and fires:

Keep ignition sources away.

Storage:

Store in a cool, dry, well ventilated place and keep container tightly closed. Keep away from sources of ignition - refrain from smoking. Keep receptacle tightly sealed.

Storage Temperature: Store at 0 - 25 °C

SECTION VIII - EXPOSURE CONTROLS/PERSONAL PROTECTION

Engineering Controls	Good general ventilation should be sufficient to control worker exposure to airborne contaminants. Ensure eyewash stations and safety showers are close to the workstation location
PERSONAL PROTECTIVE EQUIPMENT	
Eyes and Face	Avoid contact with eyes. Wear chemical splash goggles or safety glasses with a side shield
Skin	Wear impermeable gloves and suitable protective clothing. Since this product is absorbed through the skin, care must be taken to prevent skin contact and contamination of clothing
Respiratory	Protective respiratory equipment may be used based on user's own risk assessment. Full Face Mask with particle Type P2 filter.
Comments	Use a face shield, face mask, PVC gloves, and safety boots while handling

PEL: 435 mg/m³, 100 ppm

REL: Short-term value: 655 mg/m³, 150 ppm

Long-term value: 435 mg/m³, 100 ppm

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TLV: Short-term value: 651 mg/m³, 150 ppm
Long-term value: 434 mg/m³, 100 ppm
BEI

SECTION IX - PHYSICAL AND CHEMICAL PROPERTIES

Appearance	Liquid
Color	Off White
Boiling Point/ range	NA
Flash Point	> 100 °C
Density (at 20°C)	1.75 g/cm ³
Solubility In Water	Not miscible or difficult to mix

SECTION X - STABILITY AND REACTIVITY

Thermal decomposition/conditions to be avoided: No decomposition if used according to specifications.

Materials to be avoided: Strong acids

Dangerous reactions

No information

Dangerous products of decomposition:

Carbon dioxide and carbon monoxide

SECTION X - TOXICOLOGICAL INFORMATION

ACUTE EFFECTS

- Harmful if swallowed, inhaled, or absorbed through the skin
- Practically non-toxic if absorbed following acute (single) exposure.
- May cause skin irritation with prolonged or repeated contact.
- Contact with the eyes may cause moderate to severe irritation
- Ingestion may cause gastrointestinal disturbances, including irritation, nausea, vomiting, and diarrhea
- Not classifiable as to its carcinogenicity to humans.

SECTION XI - ECOLOGICAL INFORMATION

Ecological effects:

Do not allow product to reach groundwater, water course or sewage system.

SECTION XII - DISPOSAL CONSIDERATIONS

- Observe all central (Govt. of India), State, and local environmental regulations
- Must not be disposed of together with household garbage. Do not allow products to reach the sewage system

SECTION XIII - OTHER INFORMATION

The information and recommendations in literature and on labels are to the best of our knowledge reliable. However, nothing herein is to be construed as a warranty or representation. Users should make their own tests to determine the applicability of such information or the suitability of any products for their own particular purpose.

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Further information

The information in the MSDS is based on data which is considered to be accurate. However, we make no guarantees or warranty, either expressed or implied on the accuracy or completeness of this information. The conditions of handling, storage, use and disposal are beyond our control and may be beyond our knowledge. For this and other reasons, we do not assume responsibility and expressly disclaim liability for loss, damage or expense arising out of or in any way connected with the handling, storage, use or disposal of this product. This MSDS was prepared and is to be used for this product. If the product is used as a component in another product, this MSDS information may not be applicable.